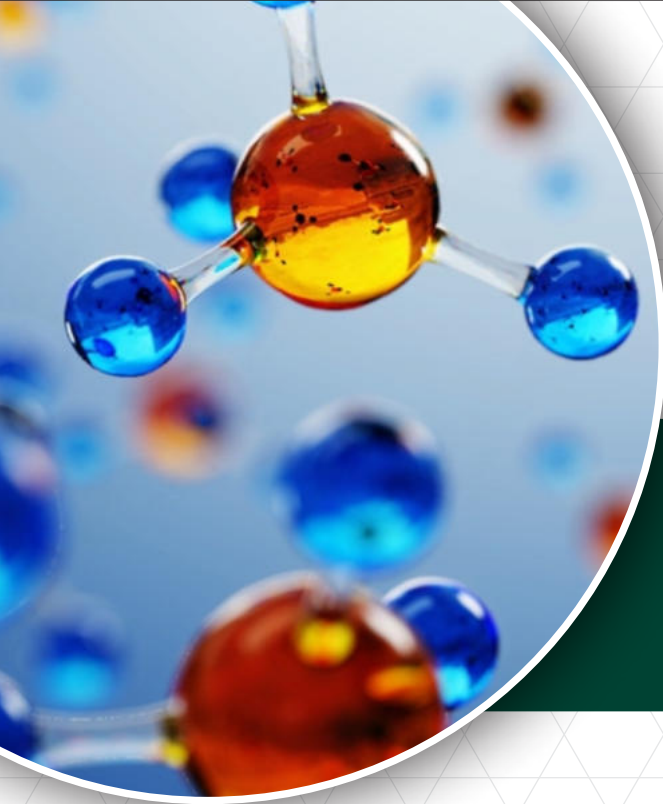
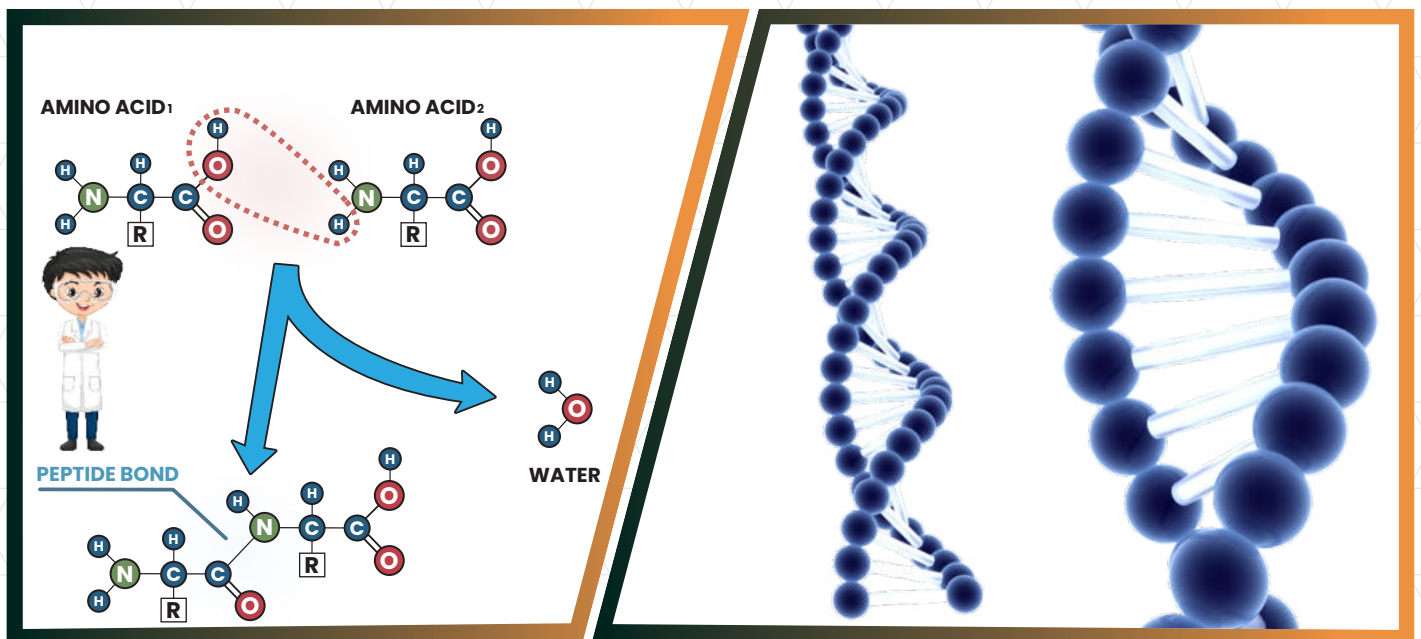




UNIT 04

BIOCHEMISTRY



WHAT IS BIOCHEMISTRY?

Biochemistry is the study of biomolecules or the chemical processes that happens inside living organisms.

CARBOHYDRATES

DEFINITION: Carbohydrates are a class of essential biomolecules found in living organisms, and they serve as one of the main sources of energy for the body.

ELEMENTS: Chemically, carbohydrates are composed of carbon, hydrogen, and oxygen atoms, with a general formula of $(CH_2O)_n$, where "n" represents the number of repeating units.

STRUCTURAL ANALYSIS

Polyhydroxy aldehydes: They contain aldehyde group ($-CHO$) with alcohol ($-OH$).

Polyhydroxy ketones: They contain ketonic group ($-CO-$) with alcohol ($-OH$).

CLASSIFICATION OF CARBOHYDRATES

Carbohydrates can be classified into three main groups based on the number of sugar units they contain:

1. MONOSACCHARIDES

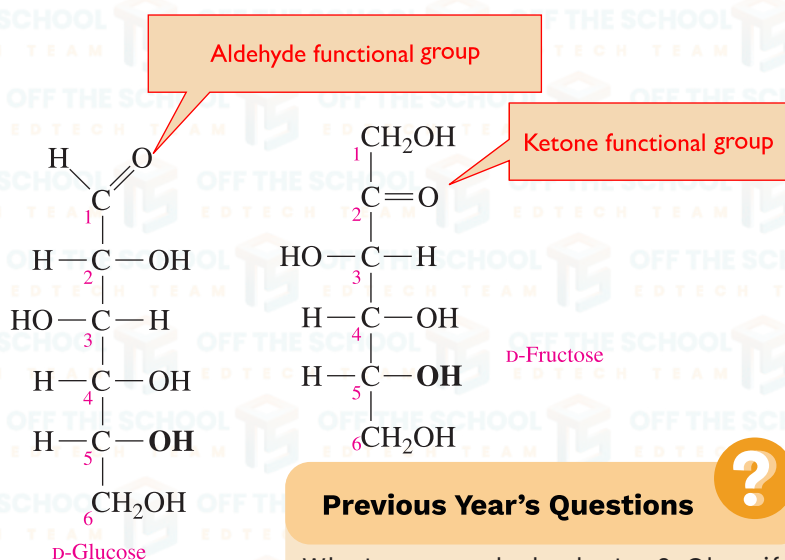
Monosaccharides, also known as simple sugars, are a type of carbohydrate that cannot be further broken down through hydrolysis.

They consist of 3 to 10 carbon atoms and can be classified into different groups such as trioses, tetroses, pentoses, and hexoses, depending on the number of carbon atoms they contain.

Monosaccharide + water $\longrightarrow$ No reaction



The sweet taste of honey is due to the monosaccharides D-glucose and D-fructose.



2. OLIGOSACCHARIDES

Oligosaccharides are a type of carbohydrate composed of a small number of monosaccharide units (typically 3 to 10) linked together through **glycosidic bonds**.

Previous Year's Questions

What are carbohydrates? Classify carbohydrates on the basis of hydrolysis into three types with an example of each also mention the source from where these can be obtained. [2024]

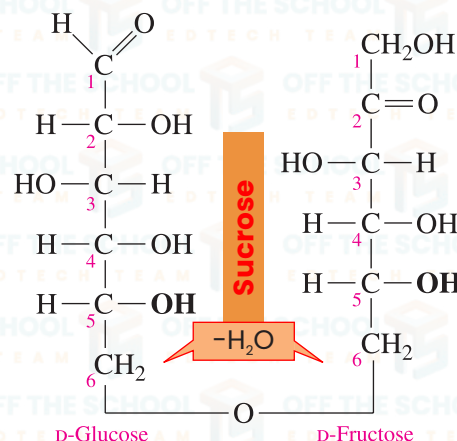
What are carbohydrates? Differentiate between aldoses and ketoses. [2025]

Oligosaccharides + water $\xrightarrow{\text{Hydrolysis}}$ **2-10 monosaccharides**



GLYCOSIDIC LINKAGE:

"The two monosaccharides are joined together by an oxide linkage formed by the loss of a water molecule. Such a linkage between two monosaccharide units through an oxygen atom is called glycosidic linkage."



3. POLYSACCHARIDES

Polysaccharides are complex carbohydrates composed of multiple monosaccharide units linked together in long chains. They serve as storage forms of energy in plants and animals and often have structural roles. Common examples of polysaccharides include starch (energy storage in plants), glycogen (energy storage in animals), and cellulose (structural component in plant cell walls).

Polysaccharides + water $\longrightarrow$ **More than 10 monosaccharides**



USES OF CARBOHYDRATES

1. Carbohydrates serve as a primary source of energy for the body.
2. Carbohydrates contribute to the structure of cells and tissues, such as cellulose in plant cell walls.
3. Carbohydrates are stored in the form of glycogen in the liver and muscles for later use.
4. Glucose, a carbohydrate, is the preferred energy source for the brain and central nervous system.
5. Carbohydrates play a role in regulating metabolism and maintaining blood glucose levels.

PROTEINS

The term "protein" is derived from the Greek word "Proteios," meaning "first." Proteins are nitrogenous macromolecules that are present in all cells of living organisms. They hold a central role in the structure and functioning of living matter. Proteins consist mainly of carbon, hydrogen, nitrogen, and oxygen, with occasional presence of Sulphur and phosphorus. A protein is defined as **a polymer, composed of simple units known as amino acids, linked together to form a larger molecule.**

AMINO ACIDS AS MONOMERS OR BUILDING BLOCKS OF PROTEIN

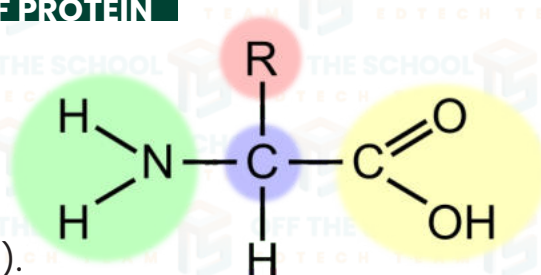
Amino acids are building blocks of proteins. They are bi-functional compounds and contain;

- ✓ Basic amino ($-\text{NH}_2$).
- ✓ Acidic carboxyl group ($-\text{COOH}$) group.

The general formula for amino acids is:

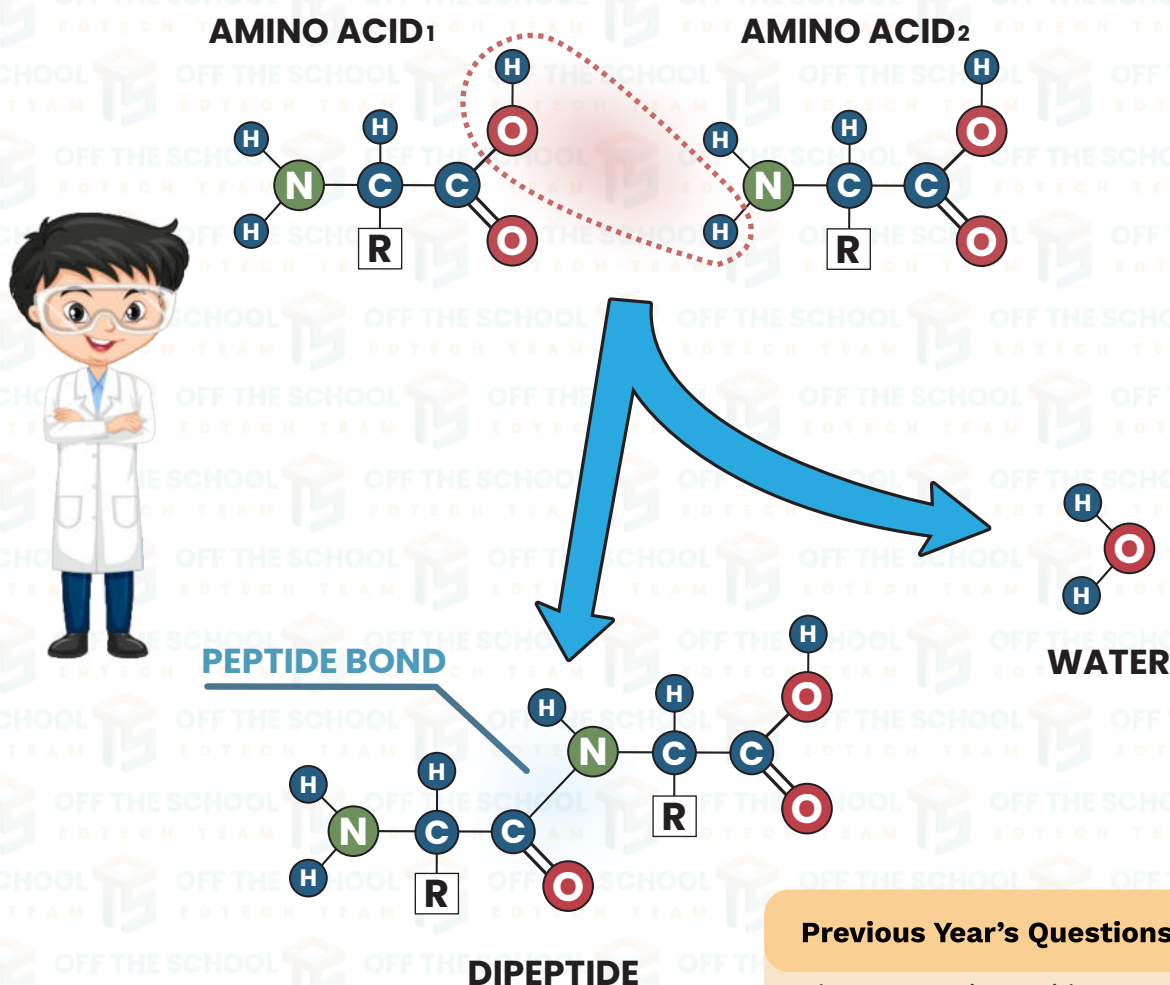
Where 'R' is the chain of carbon atoms (alkyl Radicals).

During the condensation of amino acids, $-\text{OH}$ (from carboxyl group) of one amino acid and H (from amino group) of



another amino acid are combined and eliminated as water (H_2O) molecule. Thus a new linkage is formed between two amino acid units known as **peptide linkage/bond**. Due to this linkage protein is formed.

PEPTIDE BOND



A peptide bond is a covalent chemical bond that links together two amino acids within a protein or peptide molecule.

SOURCES AND USES OF PROTEINS

The important sources of proteins are eggs, meat, pulses, beans, peas, and cheese etc.

USES OF PROTEINS

1. Proteins are crucial for building and repairing muscles.
2. Proteins act as enzymes, facilitating chemical reactions in the body.
3. Some proteins serve as hormones, regulating various physiological processes.
4. Antibodies, a type of protein, help defend the body against infections and diseases.
5. Proteins aid in the transportation of molecules, such as oxygen in the blood, and the storage of important substances like iron and calcium.

Previous Year's Questions

What are amino acids? Describe the formation of dipeptide through peptide linkage bond between two amino acid units with structural equation. [2024]

LIPIDS

Lipids are a heterogeneous group of organic compounds that are hydrophobic and soluble in organic solvents. They are essential components of cells, contributing to their structure and functioning.

FATTY ACIDS

Fatty acids are a type of organic acid that serves as a building block for lipids. They are composed of a hydrocarbon chain with a carboxyl group ($-\text{COOH}$) at one end. Fatty acids can be either saturated or unsaturated, depending on the presence or absence of double bonds in their hydrocarbon chain.

EXAMPLES:

$\text{C}_{15}\text{H}_{31}\text{COOH}$ (Palmitic Acid).

$\text{C}_{17}\text{H}_{35}\text{COOH}$ (Stearic Acid).

SOURCES AND USES OF LIPIDS

1. ANIMALS

Lipids are commonly found in animal-based foods such as meat, poultry, fish, eggs, and dairy products. Animal fats, like lard and butter, are rich sources of lipids.

2. PLANTS

Certain plant-based foods contain lipids, including oils derived from olives, sunflower seeds, soybeans, coconuts, avocados, and nuts such as almonds and walnuts.

3. FOODS

Many processed foods, such as baked goods, snacks, and fried foods, contain added fats and oils that contribute to the lipid content.

USES OF LIPIDS

1. Energy storage, providing a concentrated source of fuel for the body.
2. Structural component of cell membranes, maintaining cell integrity.
3. Insulation and protection of organs, regulating body temperature and providing cushioning.
4. Hormone production, serving as precursors for the synthesis of vital hormones.
5. Absorption of fat-soluble vitamins, facilitating their utilization in the body.

NUCLEIC ACID

The macromolecules which are formed by the polymerization of nucleotides (monomers) are called nucleic acid. They play a vital role as they store and transmit hereditary information. Nucleic acids consist of carbon, hydrogen, oxygen, nitrogen, and sometimes phosphorus.

Each nucleotide consist of:

1. Pentose sugar.
2. Phosphate group.
3. Nitrogenous base.

TYPES OF NUCLEIC ACID

There are two main types of nucleic acids:

1. DEOXYRIBONUCLEIC ACID (DNA)

DNA is a double-stranded nucleic acid that carries the genetic information in most living organisms. It is composed of nucleotides containing a sugar molecule called deoxyribose, a phosphate group, and one of four nitrogenous bases (adenine, guanine, cytosine, and thymine). The sequence of these bases forms the genetic code that

determines the traits and characteristics of an organism.

2. RIBONUCLEIC ACID (RNA)

RNA is a single-stranded nucleic acid that plays various roles in protein synthesis and gene expression. Like DNA, it is composed of nucleotides, but with a different sugar molecule called ribose. The four nitrogenous bases in RNA are adenine, guanine, cytosine, and uracil (instead of thymine in DNA). RNA molecules are involved in processes such as transcription, translation, and regulation of gene expression.

IMPORTANCE OF NUCLEIC ACID

1. Nucleic acids carry and transmit genetic information, determining an organism's traits and characteristics.
2. They play a crucial role in protein synthesis, aiding in the production of essential cellular components.
3. Nucleic acids are involved in gene expression and regulation, influencing the activity and function of genes.
4. They contribute to the development and differentiation of cells and tissues during growth and reproduction.
5. Nucleic acids are indispensable for the continuity of life, as they ensure the transfer of hereditary information from parents to offspring.

VITAMINS

Vitamins are essential organic compounds that are required in small amounts for the proper functioning and maintenance of the human body. They play crucial roles as coenzymes, helping enzymes carry out various metabolic processes.

TYPES OF VITAMINS

There are two main types of vitamins.

1. FAT SOLUBLE VITAMINS

Fat-soluble vitamins are a group of vitamins that are soluble in fats and oils, and they are absorbed and stored in the body's fatty tissues. These vitamins include vitamin A, vitamin D, vitamin E, and vitamin K. Due to their solubility in fat, they require the presence of dietary fat for proper absorption in the intestines.

2. WATER SOLUBLE VITAMINS

Water-soluble vitamins are a group of vitamins that are soluble in water and are not stored to a significant extent in the body. They include vitamin C and the B-vitamins (thiamine, riboflavin, niacin, pantothenic acid, pyridoxine, biotin, folate, and cobalamin).

Fat-Soluble Vitamins	Water-Soluble Vitamins
SOLUBILITY	
1. Fat-soluble vitamins (A, D, E, and K) dissolve in and are transported with fats and oils in the body.	Water-soluble vitamins (C and the B-complex vitamins) dissolve in water and are transported through the bloodstream.

STORAGE	
2. They can be stored in the body's fatty tissues for extended periods, which means excess intake can lead to toxicity over time.	They are not stored to a significant extent in the body, and excess intake is typically excreted in urine, reducing the risk of toxicity.
SOURCES	
3. Common sources include animal products, nuts, seeds, and oils, as well as some fruits and vegetables with high-fat content.	Found in a wide range of foods, including fruits, vegetables, grains, and legumes, as well as animal products in the case of B-complex vitamins.
ABSORPTION	
4. Fat is required for their absorption.	They are absorbed directly into the bloodstream, making them readily available for the body to use.
EXCRETION	
5. Fat-soluble vitamins are not easily excreted in urine and are eliminated from the body more slowly than water-soluble vitamins.	Water-soluble vitamins are excreted in urine if taken in excess, reducing the risk of building up toxic levels in the body.

Previous Year's Questions



Distinguish between fat soluble and water-soluble vitamin.

[2023]



**THE ONE WHO WAITS GETS ONLY
WHAT THE ONE WHO TRIES
COULDN'T GET.**

(DR. ABDUL QADEER KHAN)

